

COMPSYS302

Design: Software Practice

Introduction to AI #1 – Neural Network

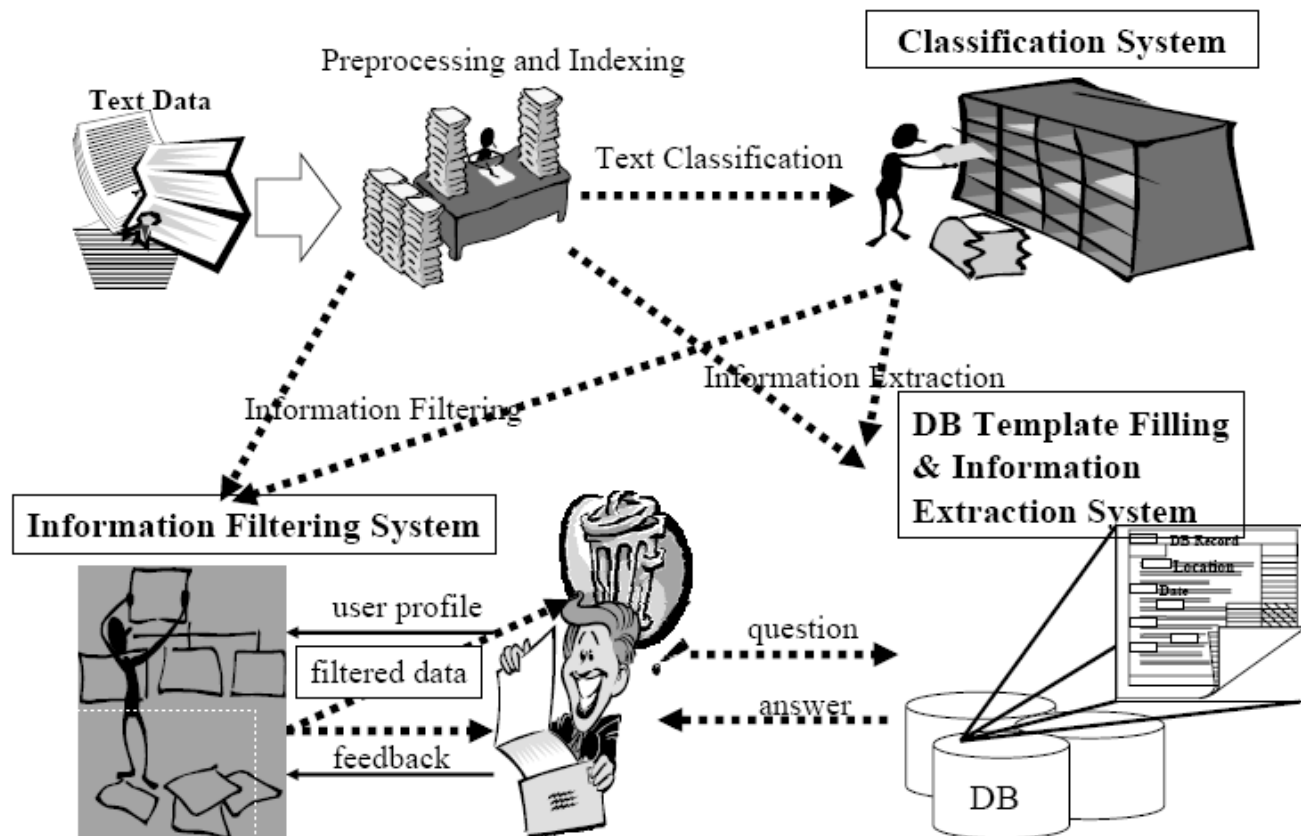
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1. Overview

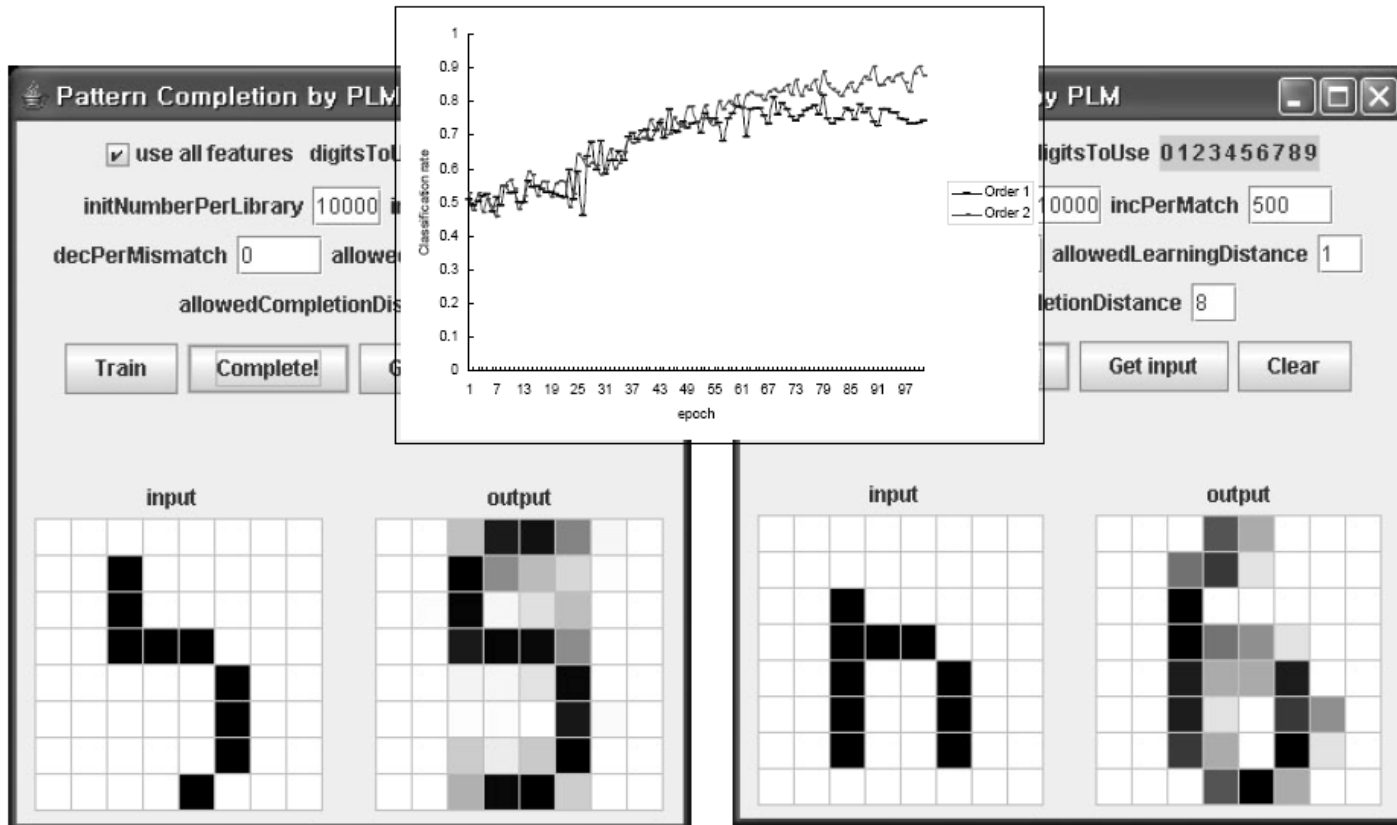
□ Where AI is used?

□ Data Analysis



1. Overview

- Where AI is used?
- Pattern analysis



1. Overview

- Where AI is used?
- Expert systems

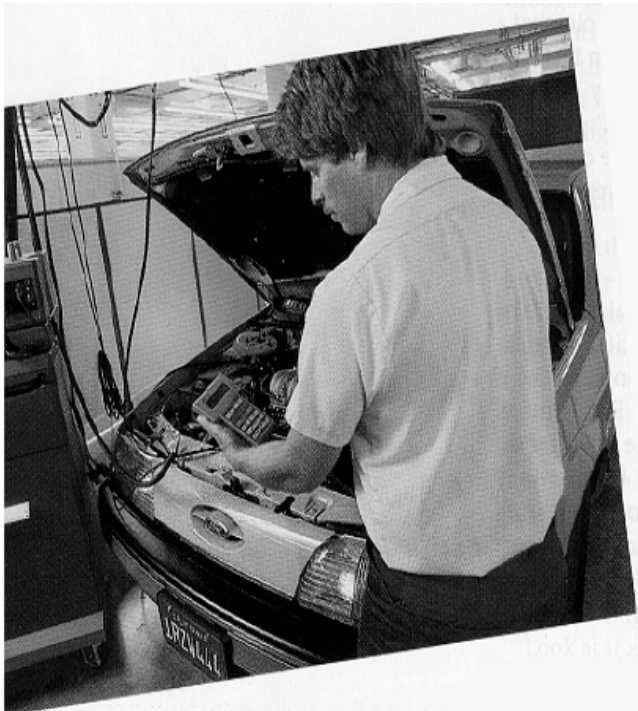


Figure 16-2 An expert system on the job. This expert system helps Ford mechanics track down and fix engine problems.

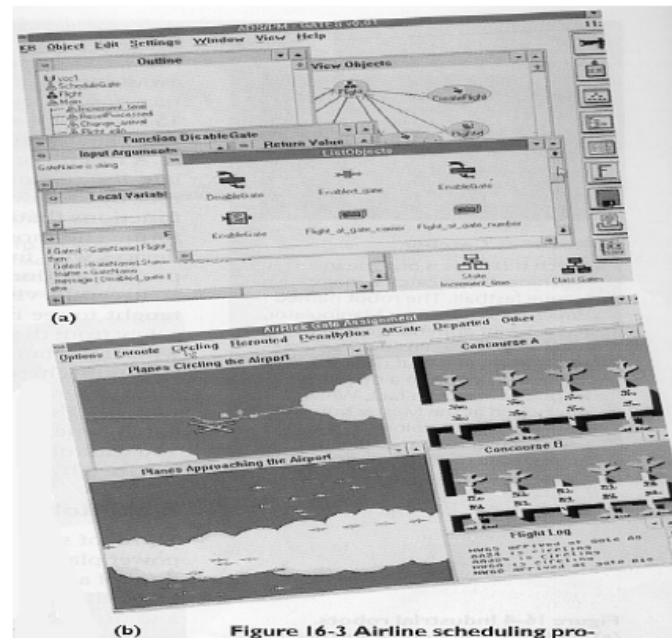


Figure 16-3 Airline scheduling program produced with the aid of an expert system. This system offers a graphical user interface to help solve a complex airport scheduling problem. (a) This screen illustrates the system's ability to display multiple views of objects and the relationships between them. (b) Various screen windows show planes circling the airport, the number of planes approaching the airport, gate information, and two concourses with planes at their gates.

1. Overview

- **Where AI is used?**

- **Robots**



1. Overview

- **Where AI is used?**

- **Robots**



1. Overview

- **Where AI is used?**

- **Robots**



NZ Robot Soccer Championship

- **When?** Mid October every year
- **Who can join?** All students who are interested in "Robot Soccer"
- **How to join?** Please contact Ho Seok AHN (hs.ahn@auckland.ac.nz)

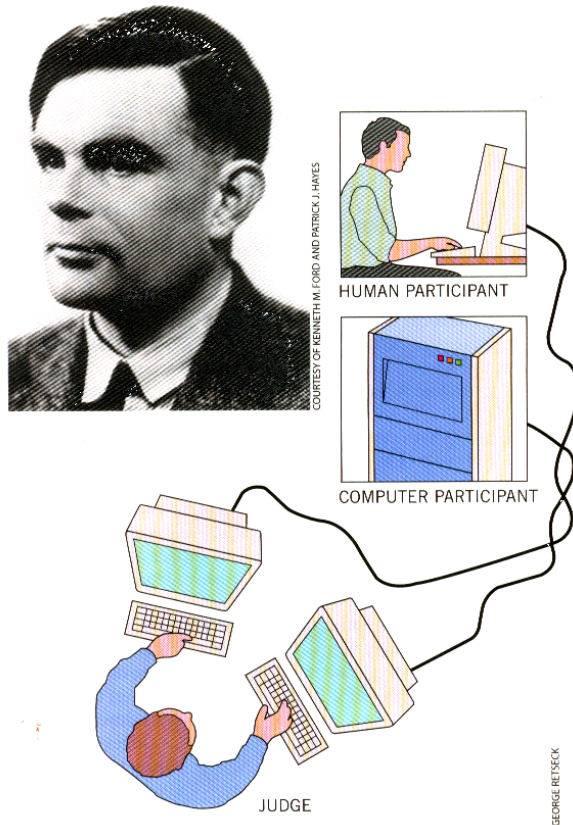
< Benefits for you >

- Having fun
- Offering you default system includes robot and vision system
- learning AI and vision processing, programming, controlling methods, system design, and cooperation
- **The winners get the IEEE certificates**
- **Can be a NZ Robot Soccer Champion**
- **Opportunities to compete with overseas teams**



2. AI machine

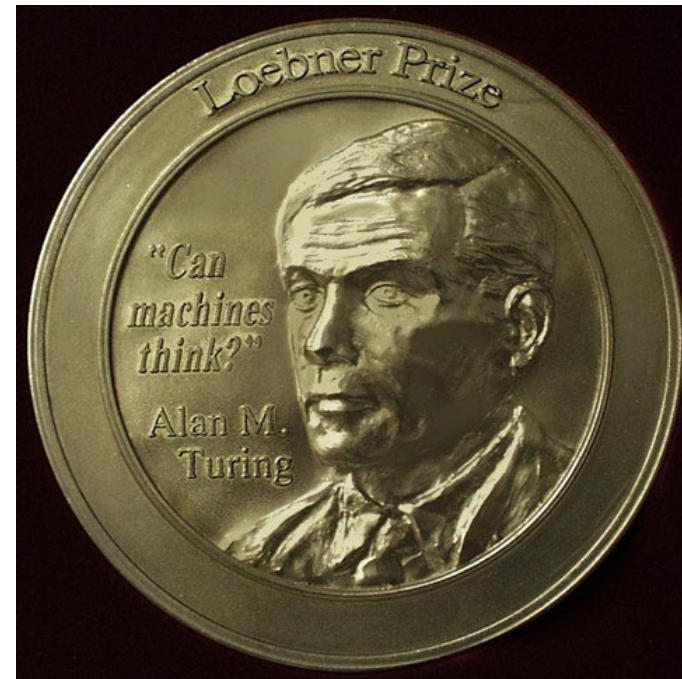
□ Can machines think?



TURING TEST for artificial intelligence was proposed in 1950 by British mathematician Alan M. Turing (photograph). In the test, a human judge would hold a three-way conversation with a computer and another human. If the judge could not distinguish between the responses of the human and those of the computer, the machine would pass the test.

The Turing Test

Computing Machinery and Intelligence [Turing, 1950]



2. AI machine

□ Can machines think?



Garry Kasparov and Deep Blue. © 1997

2. AI machine

- Can machines think?



AlphaGo vs Sedol Lee, 2016

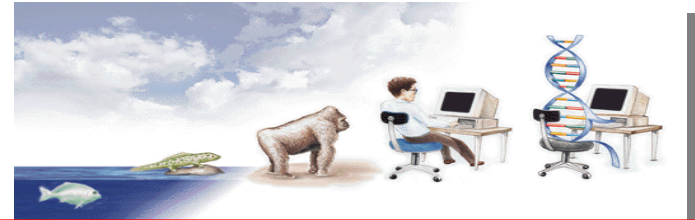
3. Types of AI

□ Paradigm

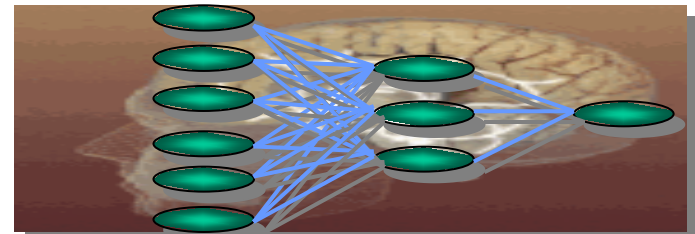
Symbolic AI
Rule-Based Systems

P	Q	R	$(P \& \neg Q) \vee R \Rightarrow (R \& \neg Q \Rightarrow \neg P)$												
T	T	T	T	F	F	T	T	T	T	F	F	F	T	T	F
T	T	F	T	F	F	T	F	F	T	F	F	F	T	T	F
T	F	T	T	T	T	F	T	T	F	T	T	T	F	F	F
T	F	F	T	T	T	F	T	F	T	F	F	T	F	T	F
F	T	T	F	F	F	F	T	T	T	T	F	F	F	T	T
F	T	F	F	F	F	T	F	F	T	F	F	F	T	T	F
F	F	T	F	F	F	T	F	T	T	T	T	T	F	T	F
F	F	F	F	F	T	F	F	F	T	F	F	T	F	T	F

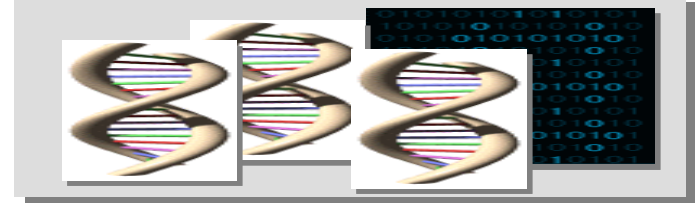
Evolutionary AI
Genetic Algorithms



Connectionist AI
Neural Networks

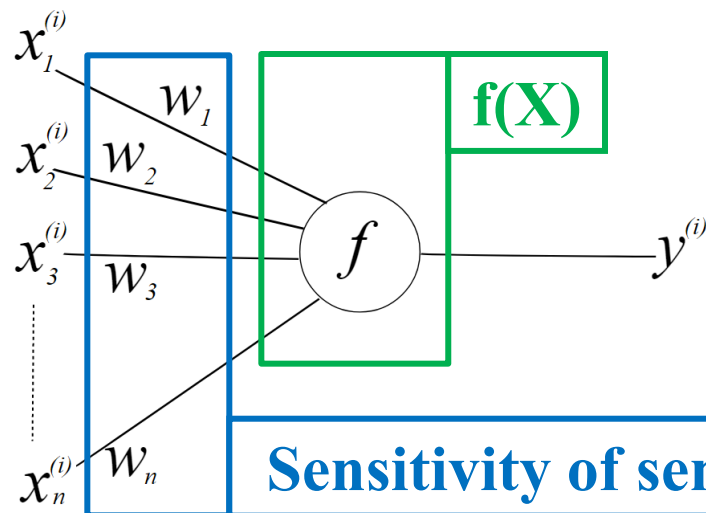
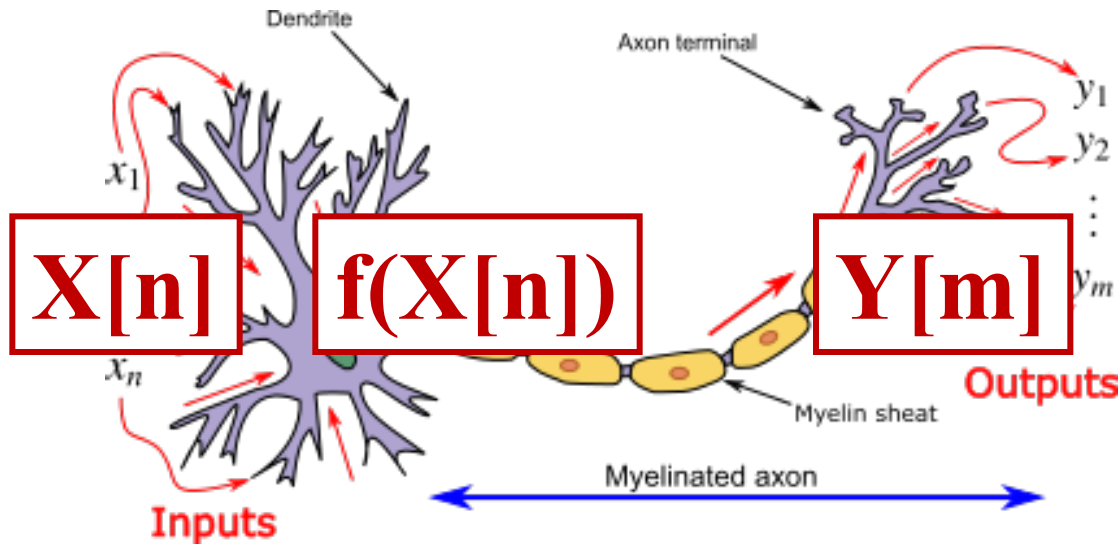


Molecular AI:
DNA Computing



4. Neural Network

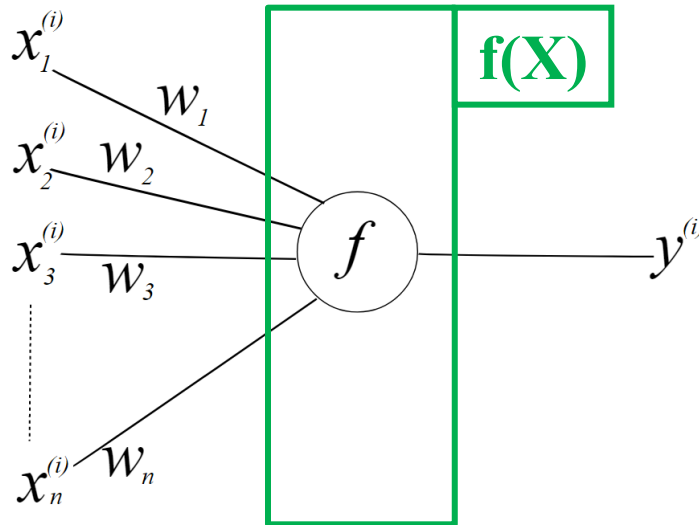
□ Neuron and Neural Network



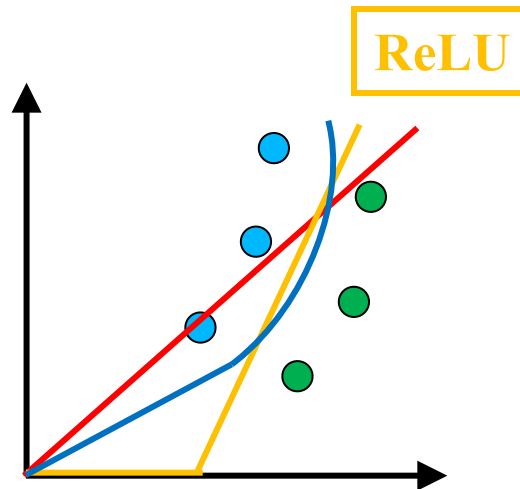
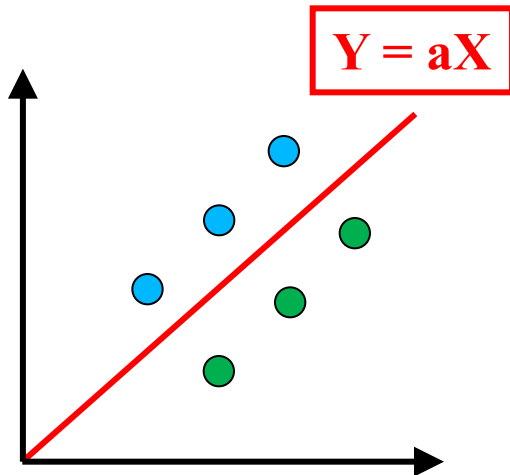
$$\text{Neural Network} \\ Y = f(wX)$$

4. Neural Network

□ Neural Network and DNN



$$\text{Neural Network}$$
$$Y = f(wX)$$



Should find $f(x)$



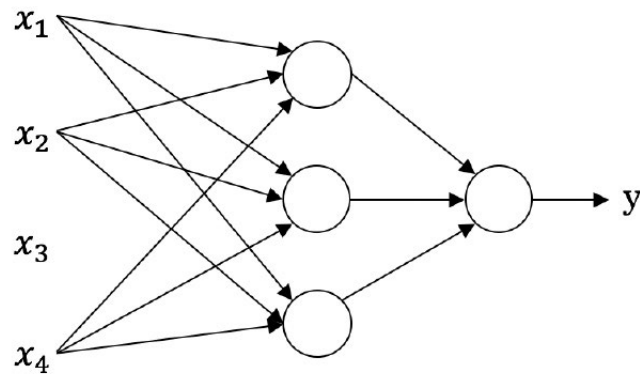
Standard NN

Convolutional NN

Recurrent NN

4. Neural Network

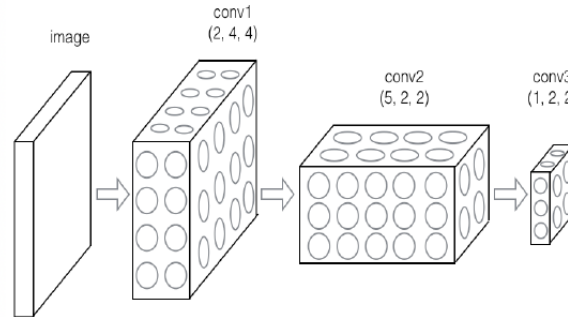
□ Types and models of NN and DNN



Standard NN

Activation function

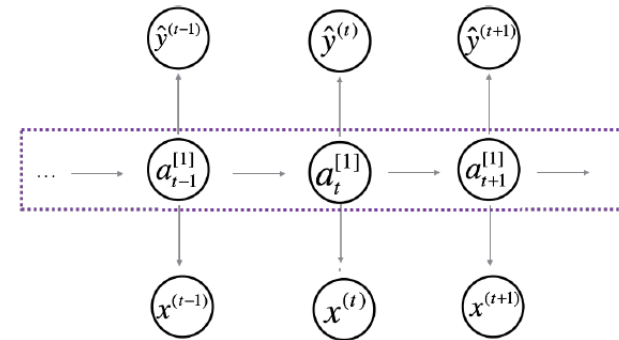
- Sigmoid()
- Tanh()
- ReLU()
- Leaky ReLU()
- ...



Convolutional NN

LeNet-5
AlexNet
ImageNet
VGG
ResNet
Inception
R-CNN

...

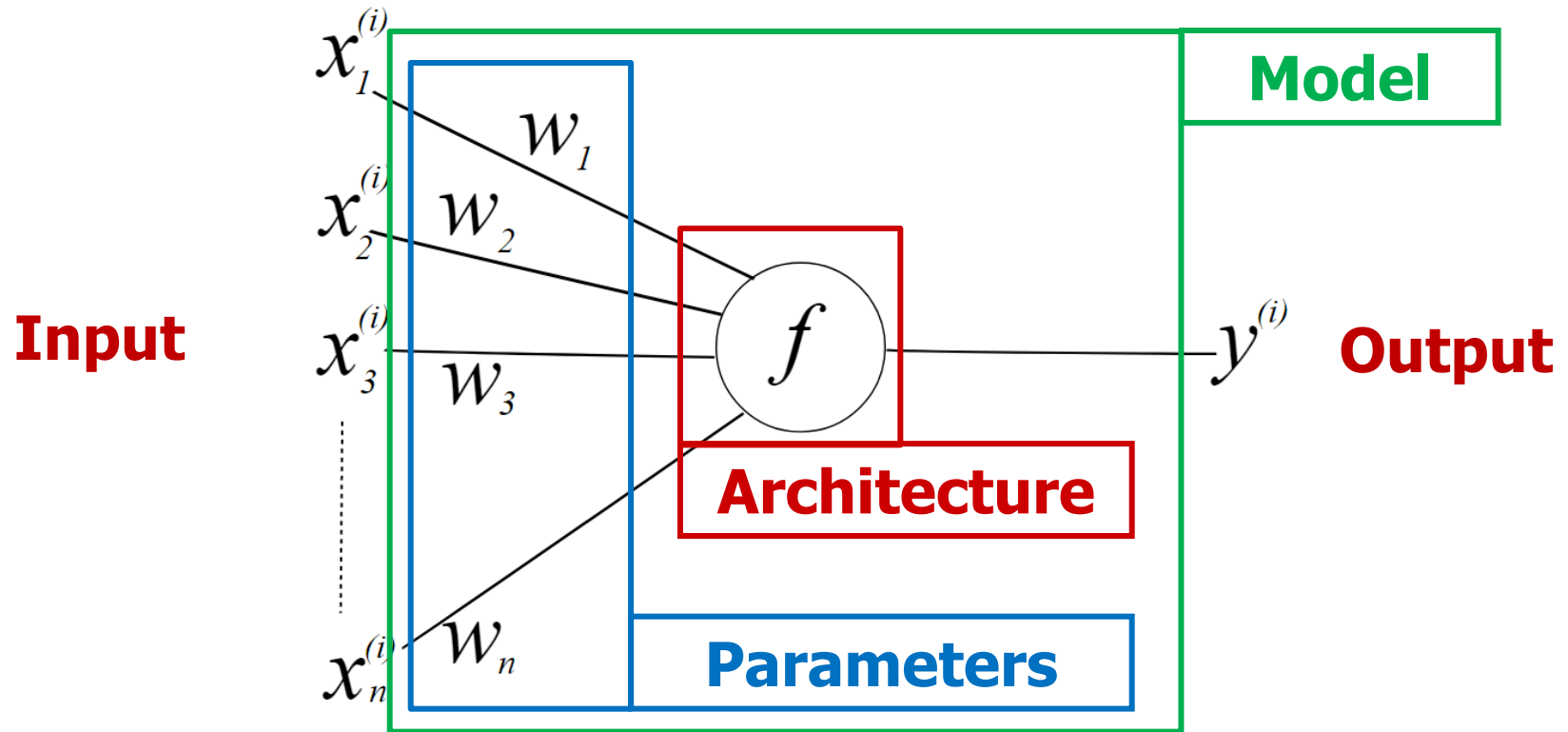


Recurrent NN

GRU
LSTM

4. Neural Network

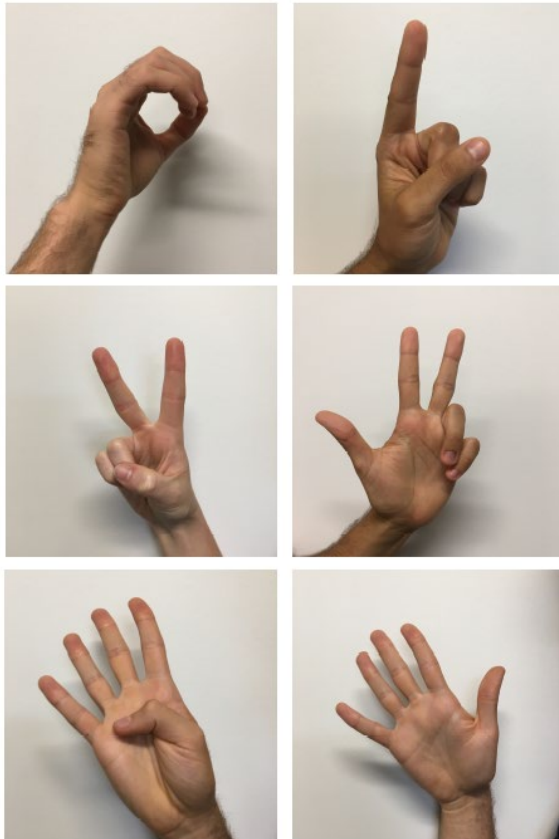
□ What is Model?



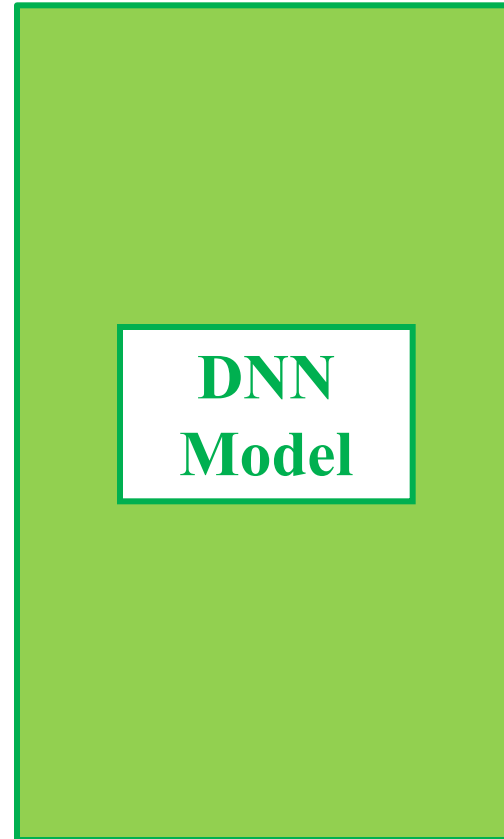
Model = Architecture + Parameters

5. Project examples

□ Sign language understanding



Input



0
1
2
3
4
5

Output

5. Project examples

□ Emotion understanding



**DNN
Model**



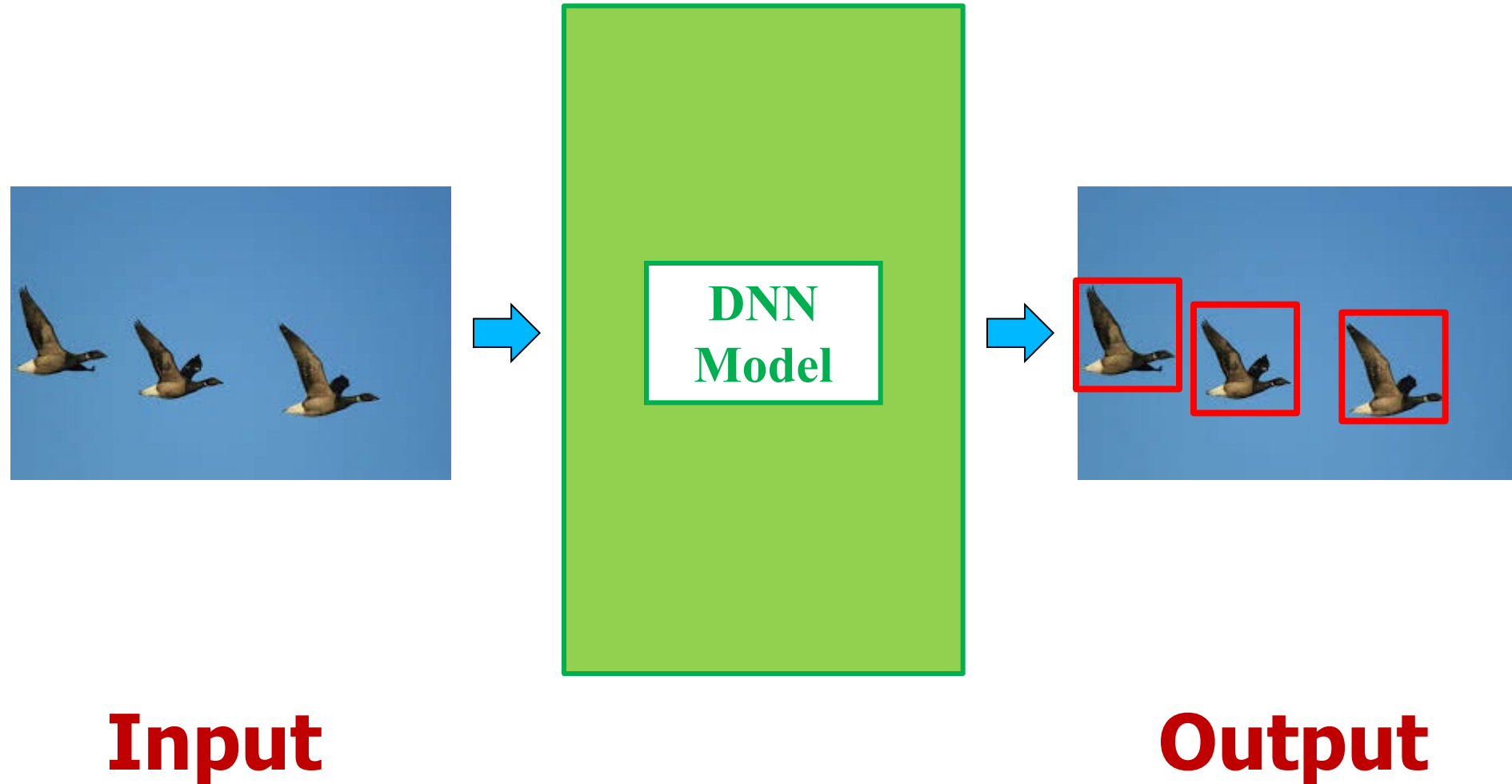
Happy
Sad
Surprise
Angry
...

Input

Output

5. Project examples

☐ Object detection



5. Project examples

□ Object recognition



**DNN
Model**



Car	0.03
Bike	0.02
Cat	0.75
Dog	0.11
Bird	0.04
...	...
Fish	0.01

Input

Output

5. Project examples

☐ Image style transfer



**DNN
Model**



In the style of Claude Monet



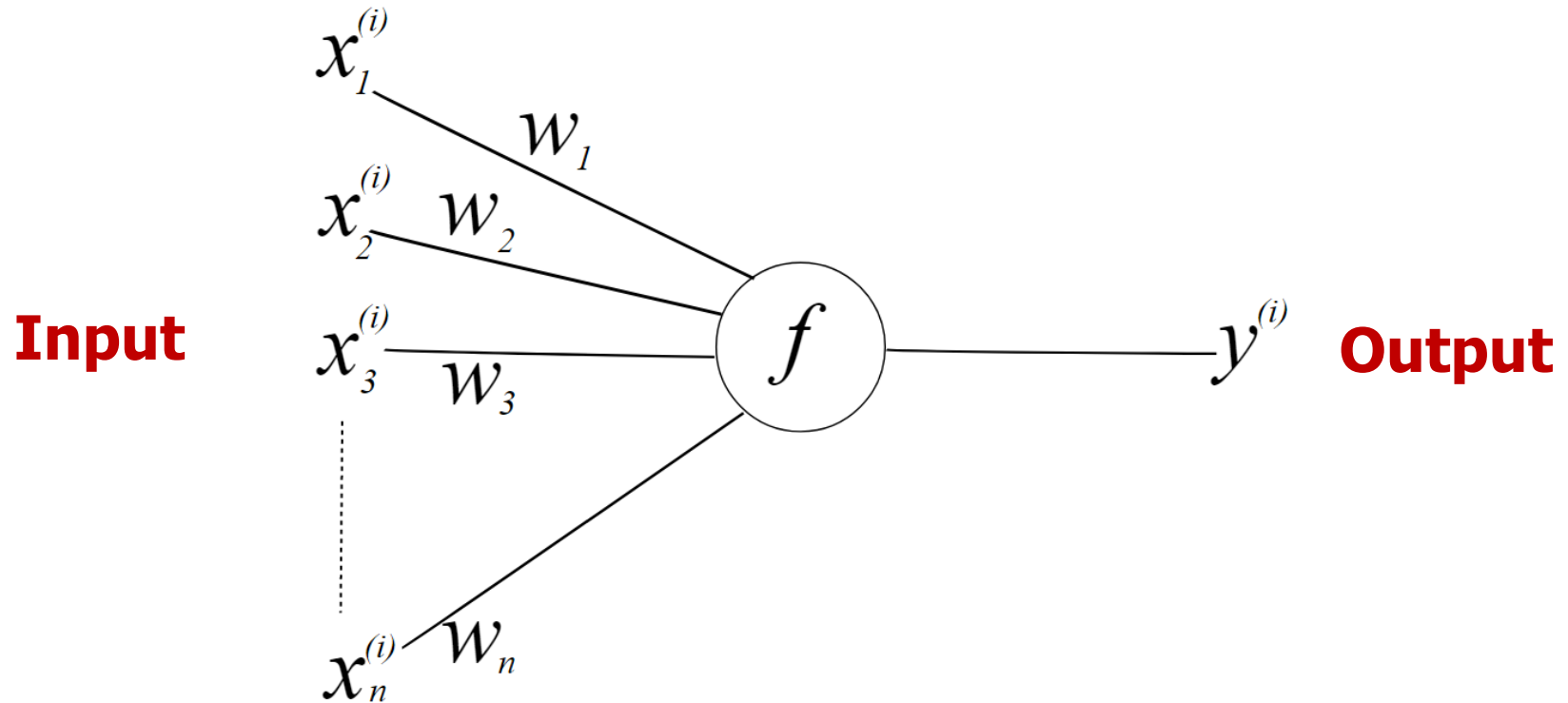
In the style of Pablo Picasso

Input

Output

6. Standard Neural Network

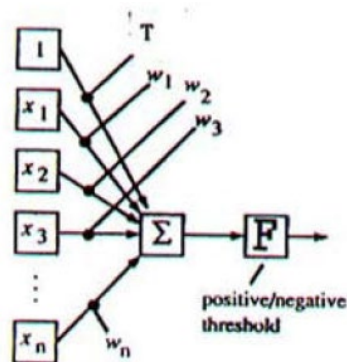
□ Neural Network



6. Standard Neural Network

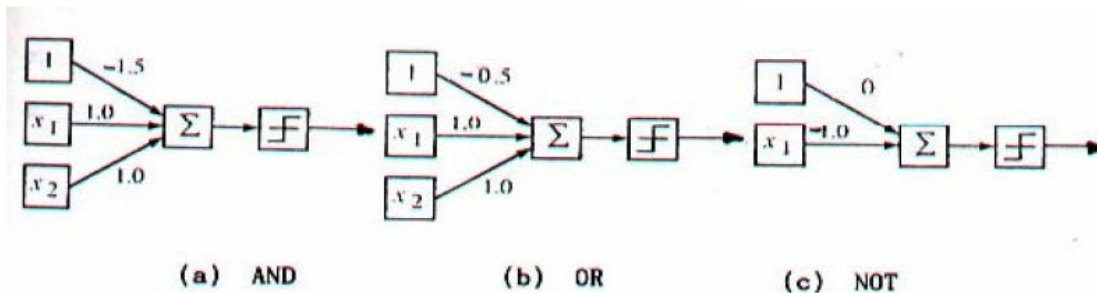
□ M-P Neuron

- In 1943, by MaCulloch & Pittz
- Paper: "A logical calculus of ideas immanent in nervous activity"
- Designed AND, OR and NOT using human's neuron model



$$\text{OUT} = F(\sum w_i x_i + T)$$

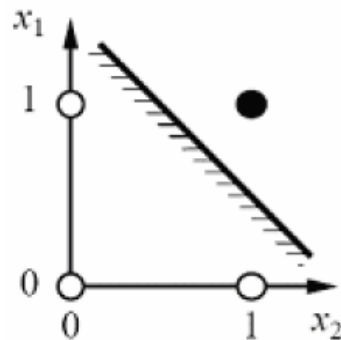
$$F(P) = \begin{cases} 1 & \text{if } P > 0 \\ 0 & \text{otherwise} \end{cases}$$



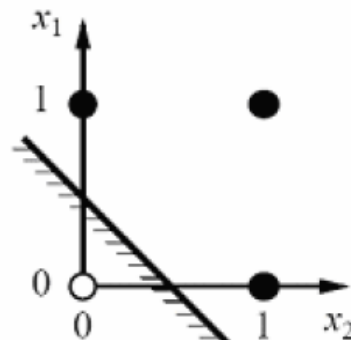
6. Standard Neural Network

□ M-P Neuron

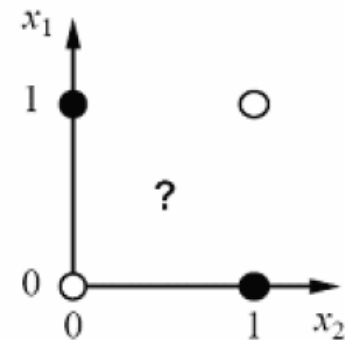
- It is good to separate two groups using linear line.
- But needs 2 linear lines for XOR.



(a) x_1 and x_2



(b) x_1 or x_2



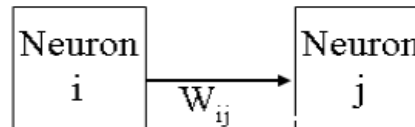
(c) x_1 xor x_2

6. Standard Neural Network

□ Hebbian Learning

- In 1949, by Hebb
- The most basic learning rule for neural network
- The **connection weight** of two neurons is the key.

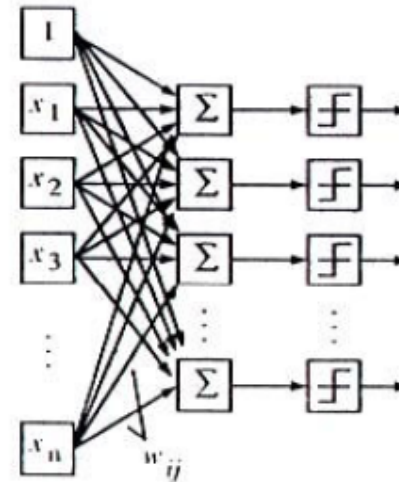
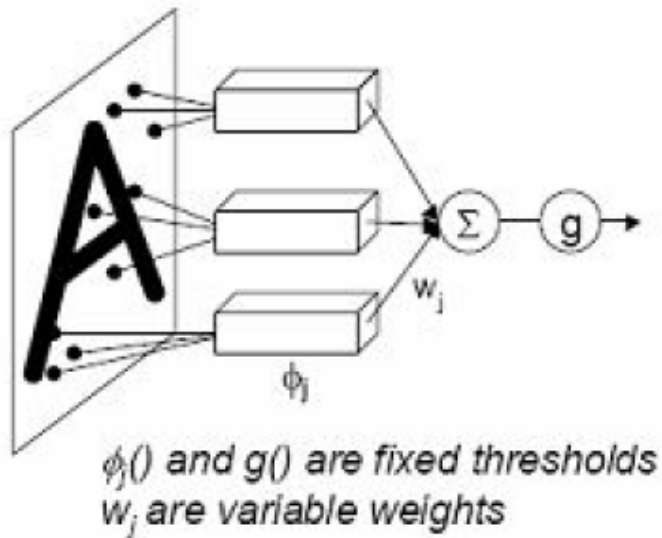
$$w_{ij}^{\text{new}} = w_{ij}^{\text{old}} + \alpha a_i b_j$$



6. Standard Neural Network

□ Perceptron

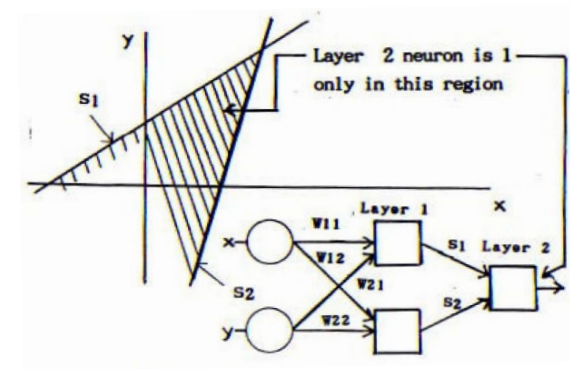
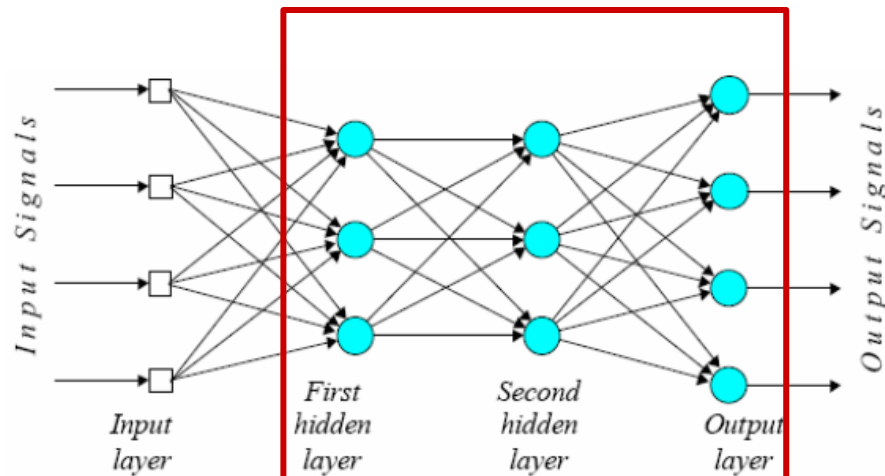
- In 1958, by Rosenblatt
- It uses **multiple neuron models** based on Hebbian Learning.
- It checks pattern.



6. Standard Neural Network

□ MLP

- We cannot solve the XOR problem with SLP (Single Layer Perceptron).
- MLP (Multilayer Perceptron) has hidden layer, and can solve the XOR problem.



If we have **a lot of layers** here...
➔ **It is DNN!!**

6. Standard Neural Network

□ Delta Rule

- In 1960, by Widrow and Hoff
- It uses the same learning rule of Perceptron.
- It updates the **weight** that **makes the minimum error** (difference) between the system output and expected value.

- Error function $\varepsilon = (d - f)^2$

- Gradient $\frac{\partial \varepsilon}{\partial \mathbf{W}} \stackrel{\text{def}}{=} \left[\frac{\partial \varepsilon}{\partial w_1}, \dots, \frac{\partial \varepsilon}{\partial w_i}, \dots, \frac{\partial \varepsilon}{\partial w_{n+1}} \right]$

- Chain Rule $\frac{\partial \varepsilon}{\partial \mathbf{W}} = \frac{\partial \varepsilon}{\partial s} \frac{\partial s}{\partial \mathbf{W}} = \frac{\partial \varepsilon}{\partial s} \mathbf{X} = -2(d - f) \frac{\partial f}{\partial s} \mathbf{X}$

- If ignore Activation Function, $f = s = \mathbf{X} \cdot \mathbf{W}$, $\frac{\partial f}{\partial s} = 1$

- Gradient $\frac{\partial \varepsilon}{\partial \mathbf{W}} = -2(d - f) \frac{\partial f}{\partial s} \mathbf{X} = -2(d - f) \mathbf{X}$

- New weight vector $\mathbf{W} \leftarrow \mathbf{W} + c(d - f) \mathbf{X}$

- Delta rule

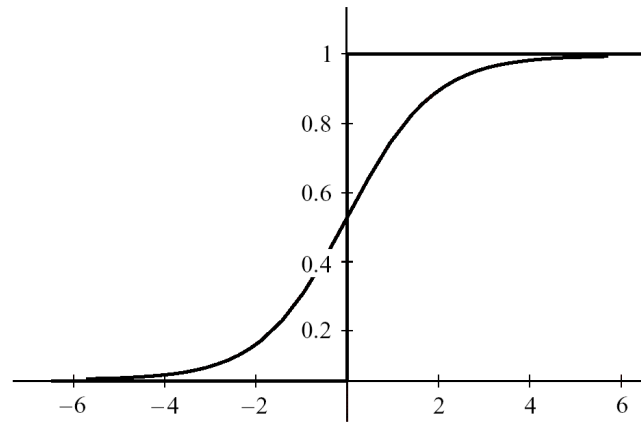
- $(d - f) > 0 \rightarrow$ increasing $s \rightarrow$ decreasing $(d - f)$

- $(d - f) < 0 \rightarrow$ decreasing $s \rightarrow$ increasing $(d - f)$

6. Standard Neural Network

□ LMS Learning Rule

- In 1986, by Rumelhart
- Least Mean Square (LMS) learning algorithm.
- We call this method as **Generalized Delta Procedure** as well.
- Use Sigmoid function as activation function.



6. Standard Neural Network

□ LMS Learning Rule

□ Error function $\varepsilon = (d - f)^2$

□ Sigmoid function (differentiable) $f(s) = 1/(1 + e^{-s})$

□ Gradient $\frac{\partial \varepsilon}{\partial \mathbf{W}} = -2(d - f) \frac{\partial f}{\partial s} \mathbf{X} = -2(d - f) f(1 - f) \mathbf{X}$

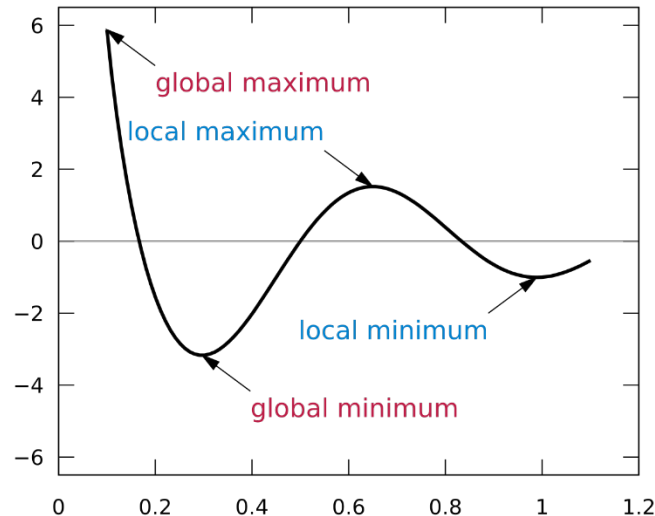
□ Generalized delta procedure

$$\mathbf{W} \leftarrow \mathbf{W} + c(d - f)f(1 - f)\mathbf{X}$$

6. Standard Neural Network

□ LMS Learning Rule

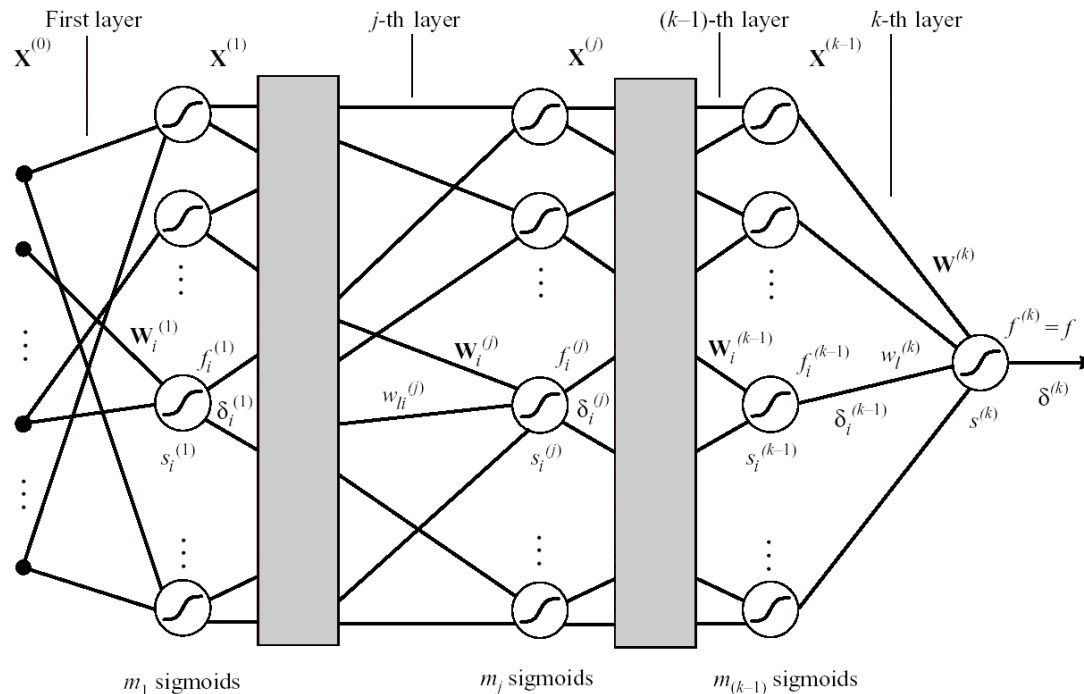
- Setup of the "Learning Rate C " is very important.
- If C is too small, local minima can be happened.
- If C is too big, it cannot find global minima.
- Usually $C < 1$



6. Standard Neural Network

□ Back Propagation

- In 1986, by Rumelhart, Hinton and Williams
- BP (Back Propagation) algorithm for MLP learning



6. Standard Neural Network

□ Back Propagation

- In 1986, by Rumelhart, Hinton and Williams

You can just use

- BP (Back Propagation) algorithm for MLP learning.

the models and functions

- Forward: get system output using input function and activation function

from existing tools,

- Backward: get error between system output and expected value.

- Learning: update the connection weights until they are smaller than the

such as

- For defined error value. Use **Generalized Delta Rule** for calculating

error.

PyTorch, TensorFlow, etc.

$$\mathbf{W}^{(k)} \leftarrow \mathbf{W}^{(k)} + c^{(k)} (d - f) f(1 - f) \mathbf{X}^{(k-1)}$$